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Coupling machine learning with a biophysical model for maturity date prediction of apple fruit across China's apple planting regions

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Abstract

BACKGROUND: Accurate prediction of apple fruit maturity date is essential for optimizing harvest timing, fruit quality and market value under climate change. However, process-based crop models often show limited performance when extrapolated across large spatial scales, whereas machine learning models lack physiological interpretability. To address these limitations, this study has developed a hybrid framework integrating the process-based STICS model with machine learning approaches across China's apple planting regions.

RESULTS: Phenological observations from 24 sites and meteorological data from 250 stations during 1991–2020 were used to calibrate and evaluate six machine learning models. Among them, the random forest (RF) model achieved the best performance [coefficient of determination (R^2) > 0.65, root mean square error (RMSE) < 8.1 days]. A hybrid approach was implemented by incorporating STICS-simulated maturity dates as input features into the machine learning models, enabling the capture of residual non-linear relationships between maturity dates of apple fruit and climatic and geographic variables. This integration further improved prediction accuracy (R^2 > 0.71, RMSE < 7.5 days), reducing errors by over 50% compared to the standalone STICS model. Spatially, the average maturity date was 282.5 ± 7.0 DOY (i.e. day of year), with the latest maturity in the Yellow River region and the earliest in the Southwest highlands. Temporally, maturity dates advanced slightly at 0.1 days decade^{-1}, with substantial regional variability. SHAP (i.e. Shapley Additive exPlanations) analysis identified chilling requirement, elevation and STICS-simulated maturity date as dominant drivers.

CONCLUSION: The hybrid STICS+RF framework effectively combines mechanistic understanding with data-driven learning, improving prediction accuracy and interpretability of apple fruit maturity date at regional scales. This approach provides a robust tool for optimizing harvest management and adapting apple production systems to climate change.
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Supporting information may be found in the online version of this article.

Keywords: autumn phenology; climate change; meteorological factors; SHAP; STICS

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INTRODUCTION

Apple (*Malus domestica* L.) is one of the most important fruit crops worldwide, with China being the largest producer and consumer.[1,2] In addition to its substantial economic value, apple has significant nutritional and health benefits, making it a cornerstone fruit in both fresh consumption and processing industries.[3-7] As perishable, high-value and seasonal commodities with time-sensitive selling prices, apple production depends heavily on the precise harvesting timing related to physiology maturity date of apple fruit, which governs yield, fruit quality, storage potential and market supply, thereby shaping outcomes for growers, processors and consumers.[8-10] Premature harvest impairs flavor and storage quality, whereas delayed harvest causes over-ripeness and disorders.[11] Thus, prediction of apple fruit maturity date is important to improve apple quality and reduce postharvest loss.[12]

Given the central role of fruit maturity date in determining apple quality and yield, it is essential to understand the factors influencing its timing, among which environmental conditions play particularly critical roles in affecting the developmental progression of apples.[13,14] Most studies have concentrated on modelling the spring phenophases of apple trees, with high accuracy at large scales,[15-22] whereas comparatively little attention has been paid to fruit maturity date. Previous studies have predicted apple fruit maturity date using process-based models such as STICS with reasonable accuracy at the site level,[15-17] as well as deep learning algorithms[23,24] and phenological models.[25] However, when extrapolating the results based on these simulation methods to regional scale, there may be significant uncertainties because many studies found that the relationship between autumn phenophases and autumn mean temperature was weaker than that between spring phenophases and spring mean temperature.[26,27] Moreover, there is still debate over whether temperature rise will advance or delay apple fruit maturity date.[28,29] These results highlighted the impact complexity of environmental factors on apple fruit maturity date.[30]

Combining crop models with machine learning has been a promising method in improving simulation accuracy and identifying the key impact factors.[23] Crop models offer mechanistic insights into crop growth, yield formation and quality development, based on physiological processes and environmental responses such as DSSAT,[31] APSIM,[32] STICS,[33,34] and so on. However, their predictive capacity is often constrained by parameter uncertainties and limitations in capturing varietal and management variabilities.[35-37] Machine learning, by contrast, excels at extracting complex, non-linear patterns from large, multi-source datasets, although it typically lacks physiological interpretability.[38,39] The integration of these approaches allows mechanistic constraints from crop models to guide machine learning, improving interpretability and robustness, whereas machine learning can enhance crop model predictions across diverse environments, management practices and climatic conditions.[40-42] This synergy enables more accurate, efficient and scalable forecasts of crop phenology, yield, quality, disease and infect pest risks, providing a solid foundation for precision agriculture and climate-resilient crop management.[38,43-47] Two commonly used approaches in crop simulation prediction involve either using process-based crop phenology model outputs directly as inputs for machine learning models to generate final predictions, or applying machine learning to estimate residual errors in process model outputs based on input variables, with the final prediction obtained by adjusting the process model output with the machine learning estimates.[46]

In this study, the apple fruit maturity date was collected across China, the world's largest apple producing country,[16,17] aiming to calibrate and validate the STICS model and machine learning model. In addition, machine learning method was used to identify the contribution of key environmental factors to apple fruit maturity date. The objectives of the study were to: (i) evaluate the performance of hybrid crop model and machine learning in simulating apple fruit maturity date; (ii) assess the spatial–temporal distribution characteristics of apple fruit maturity date during 1991–2020 across China's apple planting regions; and (iii) analyze the contribution of key environmental factors to apple fruit maturity date. These findings provide essential insights for optimizing apple harvest timing, strengthening the processing and brewing industries, and informing market positioning and policy development.

MATERIALS AND METHODS

Apple planting region

The spatial distribution of China's apple cultivation zones was mapped at $0.1° \times 0.1°$ resolution (2009–2011 average) through the Spatial Production Allocation Model (SPAM2010), as documented in multiple studies.[16,17,47,48] The study selected 250 meteorological stations within 50 km of identified apple-growing areas for nationwide phenological and quality simulations (Fig. 1). The cultivation zones were further categorized into five distinct sub-regions: (I) Loess Plateau, (II) Bohai Bay, (III) Southwest Cool Highland, (IV) Old Course of the Yellow River and (V) Specialty region. Regional distributions of principal apple cultivars are detailed in the Supporting information (Table S1).

Historical climate data

Historical climate data (1991–2020) from the China Meteorological Administration included daily records of maximum and minimum air temperatures (°C), precipitation (mm) and sunshine duration (h). Daily solar radiation (MJ m^{-2} day^{-1}) was derived using sunshine duration measurements through the Angstrom equation.[49] Spatial patterns of climatic factors during the apple growing season (April–October) during 1991–2020, as shown in the Supporting information (Fig. S1), reveal distinct latitudinal gradients across China's apple cultivation zones: both temperature and precipitation exhibited north-to-south increasing trends, whereas solar radiation demonstrated an opposite decreasing pattern from northern to southern regions.

Apple fruit maturity date simulation

Apple maturity dates were simulated using an integrated framework consisting of the process-based STICS model, machine learning approaches and their hybrid combinations. This framework enabled direct comparison of (i) the process-based STICS model, (ii) machine learning (ML) methods and (iii) the STICS+ML hybrid models, thereby providing a comprehensive evaluation of apple fruit maturity date prediction across China's staple production regions (Fig. 2).

(i) The STICS model, version 9.1[15], a process-based simulation tool driven by daily weather inputs, was applied to simulate apple fruit maturity date. Although Fuji-based cultivars dominate the dataset, cultivar-specific parameters vary across regions. The performance of the STICS model in simulating apple phenology, yield and quality in China's apple-growing regions has been validated in our previous studies, demonstrating its capability to represent perennial crop growth processes.[15,17,47] At typical 18 sites across five sub-regions, the model demonstrated good performance of apple maturity dates, with coefficient of determination (R^2)

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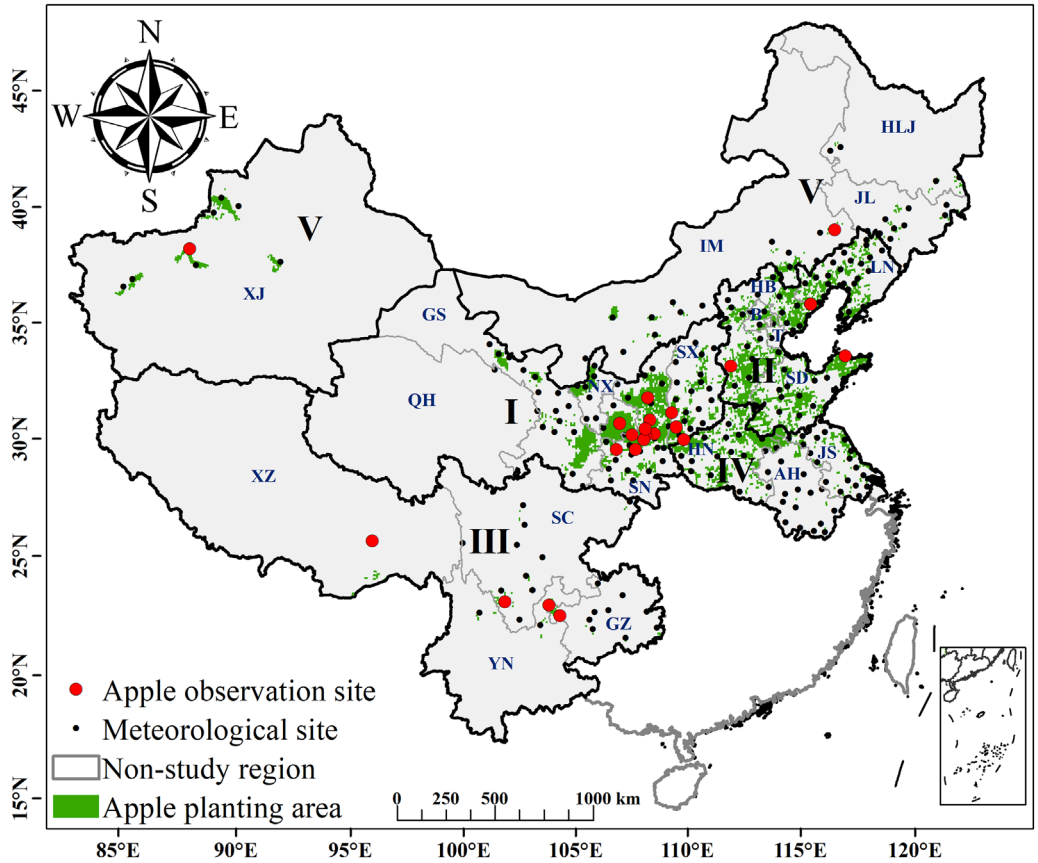


Figure 1. China's apple planting regions have five distinct sub-regions: the Loess Plateau (sub-region I), Bohai Bay (sub-region II), Southwest Cool Highland (sub-region III), Old Course of the Yellow River (sub-region IV) and Specialty Region (sub-region V). Data from meteorological stations and apple observation sites across these regions were collected for calibrating and validating the STICS and machine learning models. The geographical distribution spans multiple provinces, abbreviated as: Xinjiang (XJ), Xizang (XZ), Qinghai (QH), Gansu (GS), Ningxia (NX), Shaanxi (SN), Shanxi (SX), Hebei (HB), Beijing (B), Tianjin (T), Henan (HN), Shandong (SD), Liaoning (LN), Sichuan (SC), Guizhou (GZ) and Yunnan (YN).

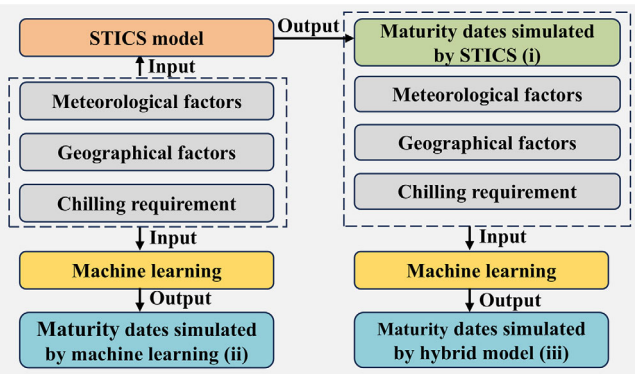


Figure 2. Flowcharts of the structure of apple fruit maturity date simulation.

> 0.70 and root mean square error (RMSE) and normalized root mean square error (NRMSE) < 9 days and 3%, respectively, between the calibration and validation groups.[17] Apple fruit maturity date was defined according to the BBCH 89 stage, at which fruits exhibit cultivar-specific characteristics in terms of size, coloration and flavor.[50] Observational data for major apple varieties (Fuji-based cultivars) were collected from 24 sites based on published literature and records from the Shaanxi Meteorological Service Station for Economic Crops (see Supporting information, Table S2). The sample sizes for maturity date of apple fruit were available across all years and sites. These data were used to calibrate and validate the STICS model. As indicated in the Supporting information (Table S3), 60% of the experimental site-years served as the calibration group, whereas the rest of the data served as the validation group. The trial-and-error method was used to derive cultivar-specific parameters by comparing the simulated and observed apple fruit maturity dates. Following calibration and validation of the STICS model, key agronomic parameters for apple cultivar-specific traits were documented in the Supporting Information (Table S4).

(ii) Climatic and geographic variables (e.g. longitude, latitude, elevation, solar radiation, heat degree days, frost degree days and chilling requirement), were then used as input features for six machine learning models: multiple linear regression (MLR),[51] support vector machine (SVM),[52] light gradient boosting machine (LightGBM),[53] eXtreme gradient boosting (XGBoost),[54] artificial neural network (ANN)[55] and random forest (RF)[56,57] to compare simulated apple fruit maturity dates. The observed data for apple fruit maturity date were pre-processed using a 70% training set and the remaining 30% were used as a test set. Hyperparameter tuning via five-fold cross-validation was performed exclusively on the training set. Moreover, the contributions of feature factors [longitude (°), latitude (°), elevation (m), total solar radiation during April–October (MJ m^{-2}), heat degree day during June–August (HDD), accumulated frost degree day during April–May (AFDD),

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Table 1. Selected environmental factors for machine learning methods

Factor	Definition	Interpretation	Unit
HDD	Heat degree days	$HDD=\sum_{Apirl}^{October} HDD_i$ $HDD_i=\begin{cases} 0, & T_{maxi}<T_{high} \\ T_{maxi}-T_{high}, & T_{maxi}\geq T_{high} \end{cases}$	°C·d
AFDD	Accumulated frost degree day	$AFDD=\sum_{df}^{dl}\mathrm{Max}(T_h-T_{min}\ ,\ 0)$	°C·d
Radn	Total solar radiation	$Radn=\sum_{Apirl}^{October} Rad_i$	MJ m^{-2}
CR	Chilling requirement	CR of different cultivars comes From Chen *et al.* (2024)[16] (see Supporting information, Table S4)	CP

Note: T_{maxi} is the daily maximum temperature for day *i*; T_{high} is defined to be 35 °C; *df* and *dl* represent apple flowering date simulated by STICS model and the last date of spring frost, respectively; T_h is 0 °C; Rad_i is daily solar radiation.

simulated fruit maturity date via STICS model and chilling requirement (CR)] were evaluated with machine learning methods with reference to Zhang *et al.*[46] Detailed environmental factors are shown in Table 1.

(iii) Simulated fruit maturity dates from STICS were further incorporated with climatic and geographic variables as additional inputs to the machine learning models. The hybrid design combined the outputs of STICS with the non-linear learning capacity of machine learning model, enabling improved prediction of apple fruit maturity dates across China's apple planting regions.

Long-term simulation setting

The simulation parameters for apple orchard configurations and genetic parameters of apple varieties, derived from the existing literature,[17] remained consistent from 1991 to 2020, aiming to focus on the impacts of changes in meteorological factors on apple fruit maturity dates.

Interpretational machine learning model on apple fruit maturity date

Shapley Additive exPlanations (SHAP) values provide local model interpretation by quantifying the contribution of each predictor variable to apple fruit maturity date, offering a detailed view of how each feature influences the model's predictions. As calculated for each observation in the training dataset, the sum of SHAP values represents the deviation of the predicted output from the field average.[58] The *shapviz* package in R (R Foundation, Vienna, Austria) was used for this analysis. Positive SHAP values indicate that a feature drives predictions toward a specific class, whereas negative values push them away.[59]

Statistical metrics for model evaluation

R^2, RMSE and NRMSE between observed and simulated apple fruit maturity dates were used to evaluate the performance of the STICS model and machine learning models:

$$R^2=\frac{\left[\sum_{i=1}^{n}(X_i-\overline{X})(Y_i-\overline{Y})\right]^2}{\sum_{i=1}^{n}(X_i-\overline{X})^2\sum_{i=1}^{n}(Y_i-\overline{Y})^2} \quad (1)$$

$$\mathrm{RMSE}=\sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i-X_i)^2}\ , \quad (2)$$

$$\mathrm{NRMSE}(\%)=\frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i-X_i)^2}}{\overline{X}}\times 100 \quad (3)$$

where X_i and Y_i represent the observed and simulated apple fruit maturity dates, respectively; $\overline{X}$ and $\overline{Y}$ represent the mean of observed and simulated apple fruit maturity dates, respectively; and *n* is the number of observation.

Spatial analysis

The spatial distribution of apple fruit maturity date was generated through inverse distance weighting interpolation, a widely utilized geostatistical approach particularly effective for regional-scale studies with moderately distributed meteorological stations.[60]

RESULTS

Performance evaluation of the machine learning model and hybrid model

Figure 3 presents the comparisons of observed and simulated apple fruit maturity dates for major apple varieties across China's apple planting regions via RF (Fig. 3(a)), XGboost (Fig. 3(b)), LightGBM (Fig. 3(c)), SVM (Fig. 3(d)), ANN (Fig. 3(e)) and MLR (Fig. 3(f)). The machine learning models demonstrated superior simulation capabilities compared to the STICS model at the provincial and site scales, with RMSE values between simulated and observed apple fruit maturity dates below 22.9 days and 12.3 days, respectively (see Supporting information, Fig. S2). For both calibration and validation periods, all machine learning models, except ANN, showed RMSE, NRMSE and R^2 values between simulated and observed apple fruit maturity dates below 10.2 days, < 3.7% and exceeding 0.44, respectively. Among the machine learning models, the RF model proved to be the most effective, with RMSE, NRMSE and R^2 values between simulated and observed apple fruit maturity dates below 8.1 days, < 2.9% and exceeding 0.65, respectively, during both calibration and validation periods.

Using the simulated apple fruit maturity date via STICS model as an input variable, Fig. 3 presents comparisons between observed and simulated apple fruit maturity dates for major apple varieties across China's apple-growing regions by machine learning models: RF (Fig. 4(a)), XGBoost (Fig. 4(b)), LightGBM (Fig. 4(c)),

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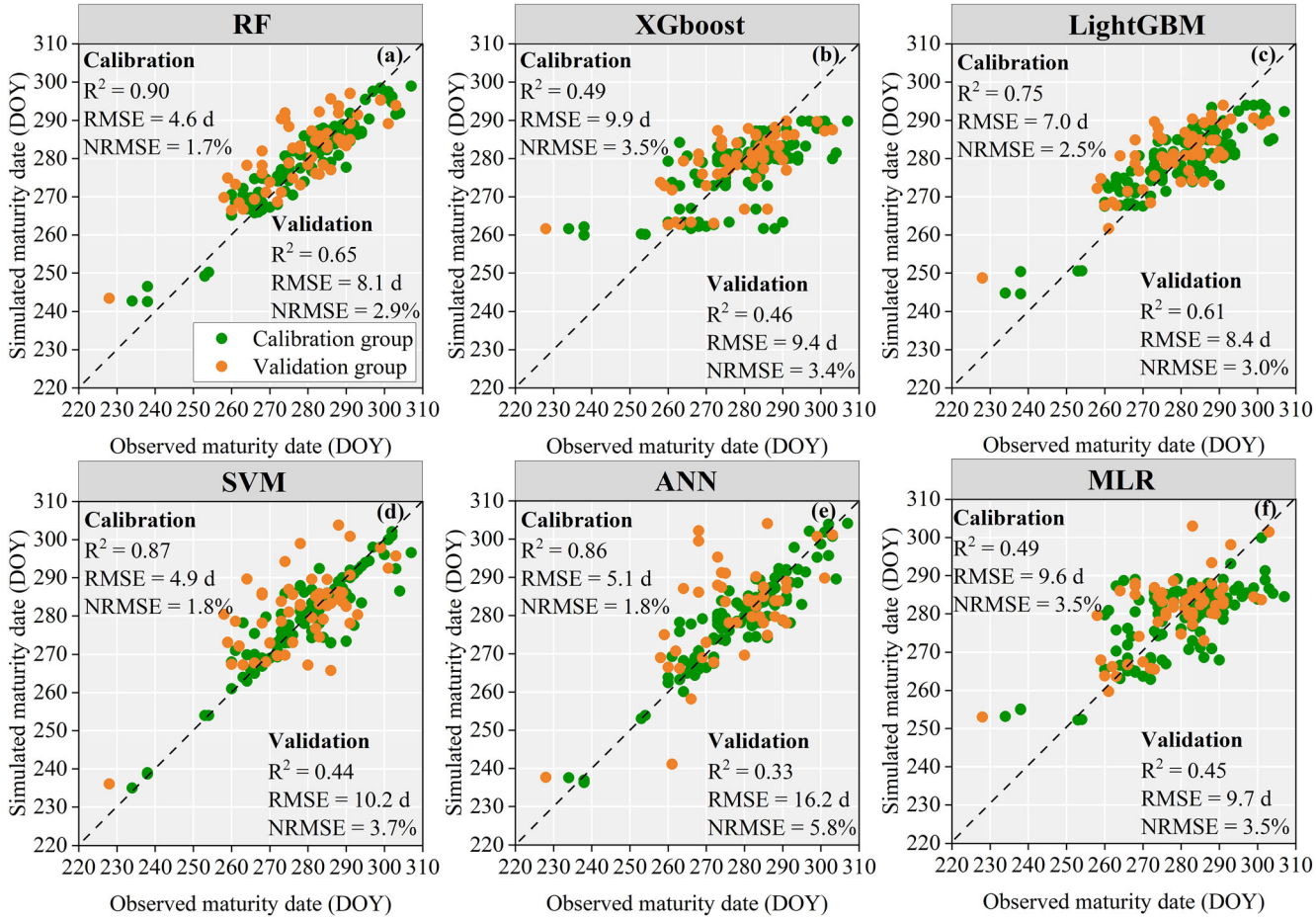


Figure 3. Observed and simulated apple fruit maturity dates based on RF (a), XGboost (b), LightGBM (c), SVM (d), ANN (e) and MRL (f) models.

SVM (Fig. 4(d)), ANN (Fig. 4(e)) and MLR (Fig. 4(f)). For both calibration and validation periods, all machine learning models, except ANN, showed RMSE ، NRMSE and R^2 values between simulated and observed apple fruit maturity dates below 10.4 days, < 3.7% and exceeding 0.40, respectively. Among the machine learning models, the RF model proved to be the most effective, and its integration with the STICS model further enhanced simulation accuracy, achieving RMSE, NRMSE and R^2 values between simulated and observed apple fruit maturity dates below 7.5 days, < 2.7% and exceeding 0.71 for apple, respectively, during both calibration and validation periods. Parameters of the random forest model are provided in the Supporting information (Table S5).

Simulated apple fruit maturity dates in China's apple planting region

Figure 5 shows the spatial distribution of apple fruit maturity date in China's apple planting region based on the STICS+RF hybrid model. Simulated date [day of year (DOY)] of apple fruit maturity date was 282.5 ± 7.0 during 1991–2020 in the entire apple planting region of China. Among all regions, the latest apple fruit maturity date (DOY) was simulated in region IV (288.3 ± 2.0), whereas the earliest occurred in region III (266.3 ± 13.2). Overall, simulated apple fruit maturity dates exhibited a quasi-longitudinal pattern, with apple fruit maturing later in eastern China than in the west. The largest difference in simulated apple fruit maturity date between sites occurred in region III across China's apple planting regions.

Figure 5 presented the temporal trend of apple fruit maturity date in China's apple planting region based on the STICS+RF hybrid model. The high-accuracy model was extrapolated into the period 1991–2020, and the results indicated that most sites exhibited no significant change in apple fruit maturity date during 1991–2020, whereas 20 sites experienced significant delay and 29 sites showed significant advancement, accounting for 8.0% and 11.6% of all sites, respectively, in Fig. 6(a). The overall apple fruit maturity date in China's apple planting region exhibited an advancing trend of 0.1 days $10a^{-1}$ (Fig. 6(b)). Regions I, II and V also showed advancement trends, with region V advancing the most at 0.4 days per decade. Moreover, regions II and IV showed delaying trends, whereas region III presented the greatest delay, with a value of 0.5 days per decade. Compared to the STICS model at the provincial scale, the STICS+RF hybrid model simulation showed a 7.0 days delay in apple fruit maturity date across China's apple planting regions (Fig. 6(c)). Simulated apple fruit maturity date showed the largest discrepancy in region IV, whereas the highest variation in the difference of simulated fruit maturity date between STICS and STICS+RF hybrid model occurred in region III.

Global and local interpretation of the key factors impacting apple fruit maturity date

SHAP interpretation of the combined STICS+RF hybrid model simulation revealed key insights into the influence of variables across China's apple planting regions during 1991–2020 (Fig. 7). The

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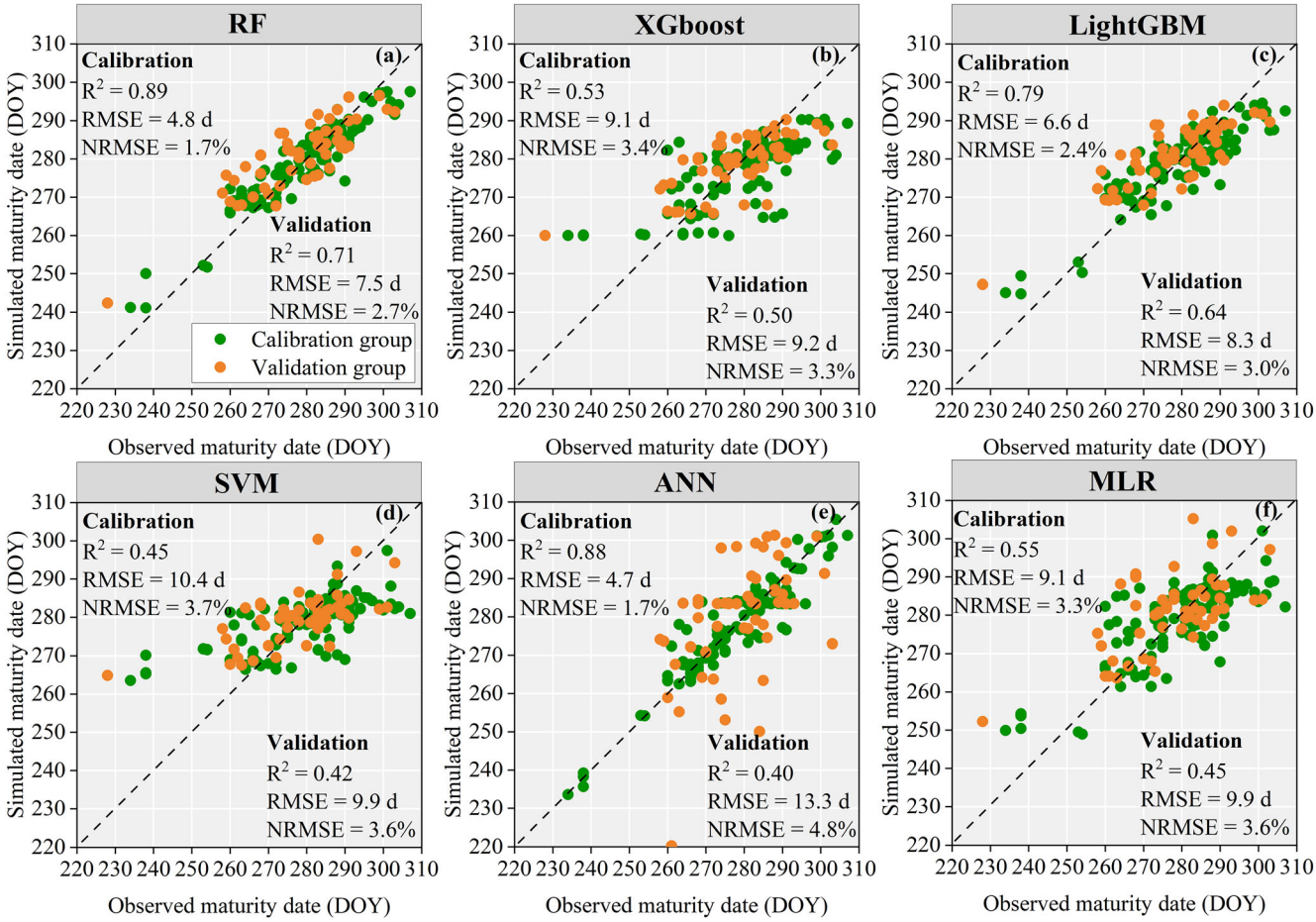


Figure 4. Comparison of observed and simulated apple fruit maturity dates from combined machine learning models [RF (a), XGBoost (b), LightGBM (c), SVM (d), ANN (e) and MRL (f)] and the STICS model.

number in the figure represent the absolute SHAP value. Chilling requirement played the most crucial role with the absolute SHAP value of 2.639. The absolute SHAP values of elevation, simulated fruit maturity date by STICS model, longitude and latitude exceeded 1.0. Total solar radiation during the apple growth season, HDD and AFDD were of relatively low importance with an absolute SHAP value below 1.0.

Local interpretation graphs (Fig. 8) revealed complex non-linear effects of individual factors on apple fruit maturity date. Longitude (Fig. 8(a)) and latitude (Fig. 8(b)) showed relatively low influence when SHAP values were low but increased sharply at specific thresholds (100–120° and 30–40°, respectively). Elevation (Fig. 8(c)), total solar radiation (Fig. 8(d)) and AFDD (Fig. 8(f)) exerted negative influences, whereas HDD (Fig. 8(e)), variety variability (CP) (Fig. 8(g)) and simulated apple fruit maturity date by the STICS model (Fig. 8(h)) demonstrated positive influences on the observed apple fruit maturity date. Among all variables, chilling requirement, elevation, simulated fruit maturity date by STICS, longitude, and latitude emerged as the dominant determinants (Fig. 7). Specifically, higher CP values corresponded to higher positive SHAP contributions, whereas elevation imposed a pronounced negative effect, with SHAP values dropping below 0 beyond 1000 m. Simulated apple fruit maturity date by STICS model displayed a consistent positive relationship, with later simulated fruit maturity date contributing more strongly to SHAP values. Later apple fruit maturity dates exerted stronger positive SHAP effects of latitude and longitude within the China's main planting region (100–120°E, 30–40°N). Total solar radiation showed a non-linear response, remaining near 0 at 3500 MJ m^{-2} but shifting to positive at lower levels. HDD maintained a strongly positive contribution with SHAP values when exceeding 0, whereas AFDD exerted a markedly negative contribution to SHAP values once above 0.

DISCUSSION

Apple fruit maturity date simulation

Process-based crop model have rarely been applied to perennial apple tree as a result of limited understanding of the underlying mechanisms affecting apple growth and development.[15,61] Our previous work demonstrated that STICS effectively captured the influence of meteorological variability on apple spring phenphases, accurately reproduced spring phenphases and consistently captured first opening flower dates for 1980–2009.[16,17] However, apple fruit maturity date in STICS was driven by thermal time from flowering to maturity, performing well at typical site scales, whereas errors exceed 20 days when the model was extrapolated to a broader region (see Supporting information, Fig. S1). In this study, combining machine learning and crop model can significantly enhance the simulation accuracy of apple fruit maturity date than using crop model or machine learning method only (Figs 1 and 2; see also Supporting information,

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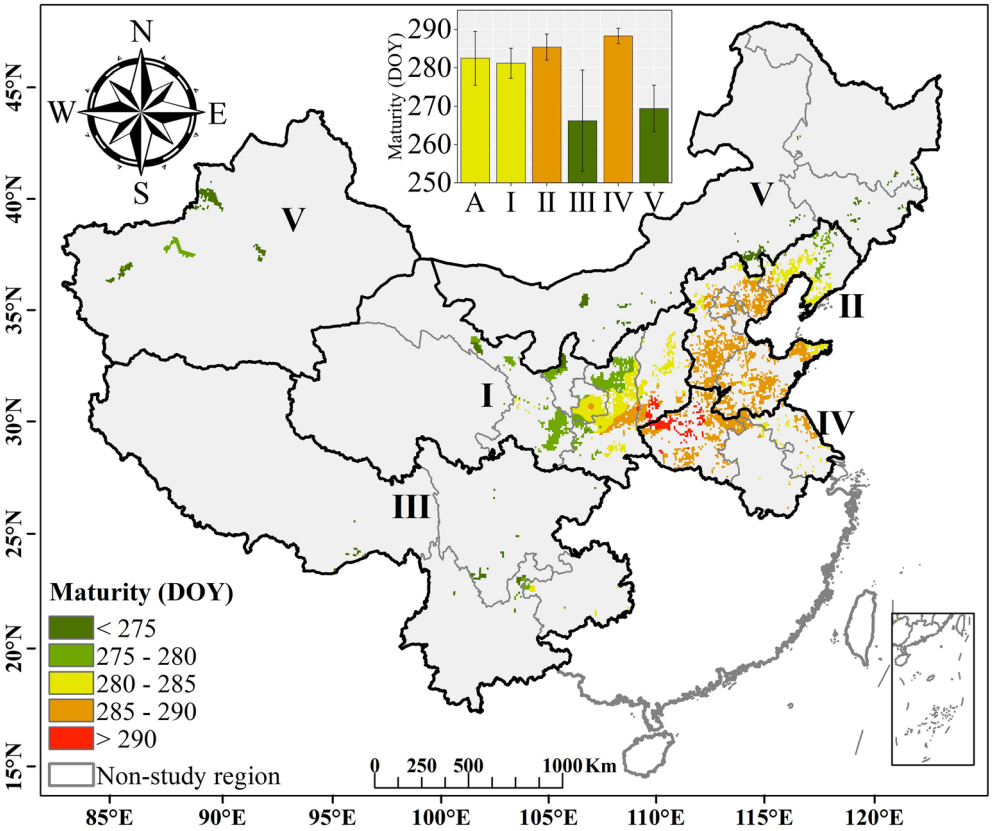


Figure 5. Simulated apple fruit maturity date with STICS+RF hybrid model in China's apple planting region during 1991–2020. The insets in Fig. 5 display simulations of apple fruit maturity date, showing both composite averages from all growing regions (A) and individual region values. Error bars represent the SD calculated within respective subregions.

Fig. S1). Integrating crop model outputs into machine learning yields superior predictive performance compared to using machine learning with respect to improving phenology-based decision support at the same time as preserving interpretability.[30,46,54] Apple maturity date was a critical indicator of quality formation, reflecting the integrated response of physicochemical properties (e.g. sugar–acid ratio, firmness and coloration) to environmental conditions.[62] Accurate prediction of the maturity date provides a scientific basis for optimizing postharvest storage potential and processing suitability for products such as juice and cider, helping to maintain fruit firmness and minimize postharvest losses.[63] Future research will further couple quality dynamic models to enhance the model's guiding value across the entire industrial chain.

The hybrid STICS+ML framework provided a practical tool for operational apple maturity prediction, supporting harvest scheduling by identifying optimal picking windows and reducing risks of premature or over-mature harvest, while preserving key quality attributes such as sugar–acid balance, firmness and coloration.

Moreover, integrating machine learning with crop models has proven effective for simulating spatiotemporal variation and parameter estimation in crop phenology and yield.[38,43,44] Recent advances in modeling suggest that empirical approaches may substitute certain submodules of process-based models when mechanistic descriptions are limited.[46,64]

Among the machine learning methods, the RF model exhibited the best performance, highlighting its robustness for crop phenology simualtion.[61] This further demonstrates the effectiveness and transferability of RF model in predicting crop traits.[65] The effects of key factors on apple fruit maturity date may be non-linear, which can result in suboptimal performance of MLR model.[57] The relatively poor performance of ANN could often be attributed to insufficient training data and limited hyperparameter optimization.[66] Although hyperparameters were tuned using five-fold cross-validation in the study, the limited observation data were insufficient to fully exploit the non-linear learning capacity of ANN, leading to suboptimal generalization. Future work will explore larger datasets and more advanced architectures to better leverage ANN capabilities. SVM performance is highly sensitive to kernel selection and parameter tuning, which may lead to suboptimal results under noisy data conditions.[67] Although LightGBM and XGBoost achieve relatively high simulation accuracy, they require more hyperparameters and incur greater computational costs compared to RF.[68] Therefore, the evaluation of machine learning models should be performed through a comprehensive assessment process that considers multiple models.[41]

Apple fruit maturity date response to climate change

Under climate warming, autumn phenology in apple fruit demonstrated significant sensitivity and complexity.[69] Overall, climate change generally advanced the physiological maturity date in terms of quality components in perennial fruit crops such as grapes and apples.[70,71] In this study, the overall tendency of apple fruit maturity date across the China's apple planting regions was a slight advancement by about 0.1 days per decade, consistent with general evidence that warming accelerated the completion

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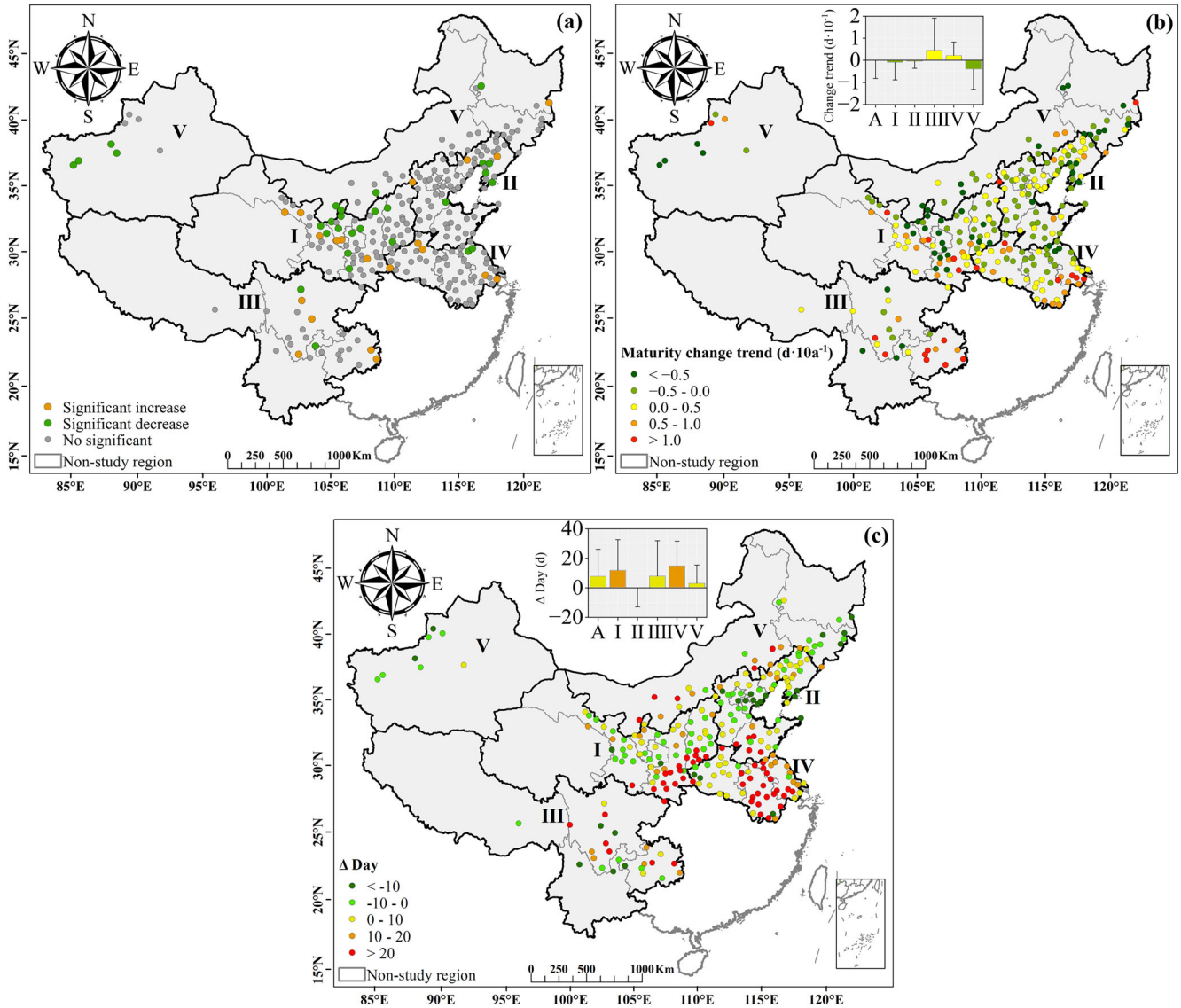


Figure 6. Simulated change in apple fruit maturity date with the significance (a), trend (b) and differences in simulated fruit maturity date (c) between STICS and STICS+RF hybrid model in China's apple planting region during 1991–2020. The insets in Fig. 6(b), (c) show the change trend in apple fruit maturity date and differences in simulated fruit maturity date between STICS and STICS+RF hybrid model averaged from all the regions (A) and in each region (I—V).

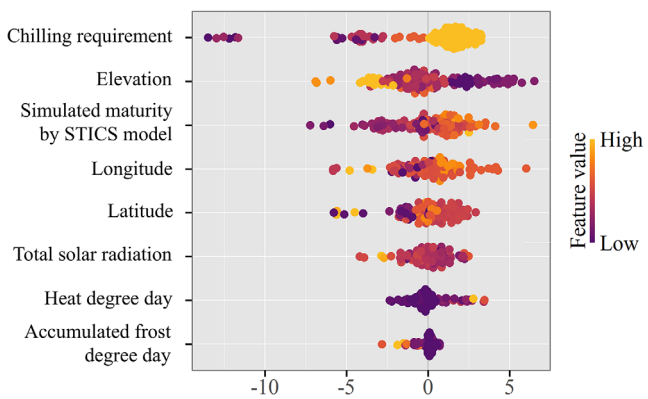


Figure 7. Global interpretation based on SHAP. Contribution rate of each factor to simulated apple fruit maturity date by the STICS+RF hybrid model.

of fruit development.[72] However, our study simulated the delayed apple fruit maturity date in Linzhi, Wanrong and Luochuan, which was consistent with previous findings.[27,28] The delayed apple fruit maturity could be primarily attributed to high chilling requirement cultivars and the cooler high altitude climate. In Akesu, Xinjiang, simulated result showed an opposite trend to observation, although the smaller advanced trend simulated by the STICS+RF hybrid model (1.9 days per decade) than that simulated by the STICS model (10.5 days per decade). This discrepancy is mainly attributable to differences in the temporal coverage of the datasets. This may be attributable to lack more data and its higher exposure to frost risks and limitations in geographic information data,[16,17] which needs to be further investigated.

In addition, accurate assessment of the apple fruit maturity date is critical for both yield composition and quality formation in apples.[62] Our previous study showed that climate change has led to relative yield increases in China's apple planting region.[17]

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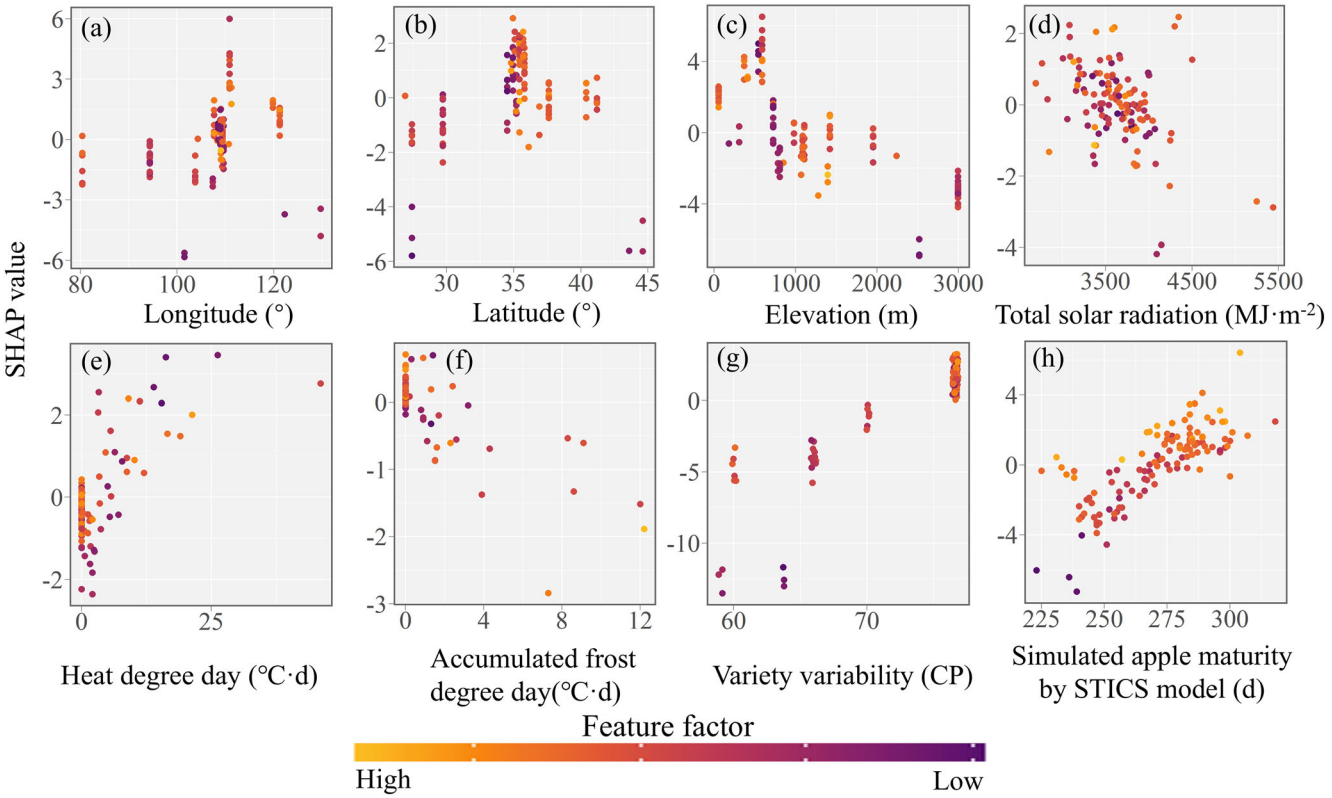


Figure 8. Local interpretation based on SHAP: Contribution rate of each factor to simulated apple fruit maturity date by the STICS+RF hybird model. Fig. 8a–h represented longitude, latitude, elevation, total solar radiation, heat degree day, accumulated degree day, variety variablity, and simulated apple maturity by STICS model, respectively.

In this study, simulated apple fruit maturity date with the hybrid model was found to be generally delayed compared with STICS simulations, suggesting that under future climate change scenarios, projected apple yield could increase even further.

Interpretation of the key factors impacting apple fruit maturity date

High-chilling requirement cultivars typically required both greater chilling accumulation and higher heat units, which delayed spring phenophases and prolonged subsequent fruit development. Consequently, the entire phenological sequence was shifted later in the season, with the extended developmental trajectory ultimately resulting in later maturity.[73] High elevation sites were typically characterized by lower mean temperature and larger diurnal temperature fluctuation, which contributes to delayed apple flowering and prolonged developmental periods.[74] The STICS model simulated apple fruit maturity date based on thermal time,[33,34] which resulted in substantial errors in warmer regions (i.e. regions I and IV), whereas the simulations were relatively consistent with observations in cooler regions such as region II. The positive contribution of latitude and longitude to model interpretability is mainly concentrated between 90°E and 120°E and 35°N and 40°N, corresponding to the China's apple staple planting regions I, II and northern parts of region IV.[47] These areas are dominated by high-chilling requirement cultivars of the Fuji-based, which are characterized by later fruit maturity date. Solar radiation plays a crucial role in apple fruit maturity date because higher solar radiation enhances photosynthesis, sugar accumulation and skin coloration, thereby accelerating fruit development, whereas lower solar radiation often delayed apple fruit maturity date and compromised fruit quality.[75] Extreme heat stress typically delays fruit maturity date by inhibiting key physiological and biochemical processes involved in maturation, including anthocyanin synthesis and sugar accumulation at the same time as impairing the photosynthetic apparatus in leaves, reducing energy supply and thereby prolonging the overall maturity process.[76] The greater the frequency or severity of frosts, the higher the likelihood of prematurely terminating or shortening the growth period, thereby advancing the apple fruit maturitydate.[22,77]

In addition, data on apple quality indicators were relatively difficult to obtain, and incorporating quality data simulated by STICS model did not improve the performance of any model (see Supporting information, Table S6).

Study limitations

Our study had several limitations. First, limited data of apple fruit maturity date, combined with inherent observational errors, diverse applications for fresh consumption and processing, collectively contribute to considerable uncertainty in assessing the spatio-temporal characteristics of apple fruit maturity date. Second, apple quality provides a direct indicator of fruit maturity date by representing sugar accumulation, acid degradation, and firmness loss, changes in shape sweetness, acidity, and texture, thereby guiding the determination of optimal harvest timing.[63,78] Therefore, future studies should consider the role of apple quality in impacting apple fruit maturity date. Third, this study did not consider the differences in harvest standards and processing uses of apples at different regions.

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Finally, our current methodology did not consider the impact of agronomic management practices (e.g. rootstock choice, irrigation scheduling, pruning regimes, and the use of reflective mulches and growth regulators) on apple fruit maturity date due to limited and inconsistent data availability across sites.

CONCLUSIONS

By integrating mechanistic crop modeling with a machine learning model, this study achieved a significant improvement in predicting apple fruit maturity date in China's apple planting region. The STICS+RF hybrid model reduced the simulation errors to below 7.5 days, and captured spatio-temporal variability of apple fruit maturity date. The STICS+RF hybrid model revealed key determinants of apple fruit maturity date, with chilling requirement and elevation emerging as dominant drivers, whereas solar radiation, heat degree days and accumulated frost days exerted secondary yet non-linear influences. Simulated apple fruit maturity date showed a slight advancing trend of 0.1 days $10a^{-1}$ during 1991–2020 in the entire apple planting region of China, although with substantial regional heterogeneity. These results highlight the value of hybrid modeling in improving fruit maturity date prediction, offering insights for growers with respect to harvest scheduling and reducing post-harvest losses, and for policymakers with respect to designing adaptive strategies to climate change. In the future research, the mechanistic understanding of apple fruit maturity date should be strengthened for advancing process-based model development.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

AUTHOR CONTRIBUTIONS

RC and JW were responsible for reviewing and editing. RC, JW, ZG, JZ, YL, RB and ZL were responsible for methodology. RC was responsible for writing the original draft, formal analysis and data processing. JW and ZG were responsible for supervision. JW, ZG, JZ, YL, RB and ZL were responsible for conceptualization. ZQ was responsible for resources and funding.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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